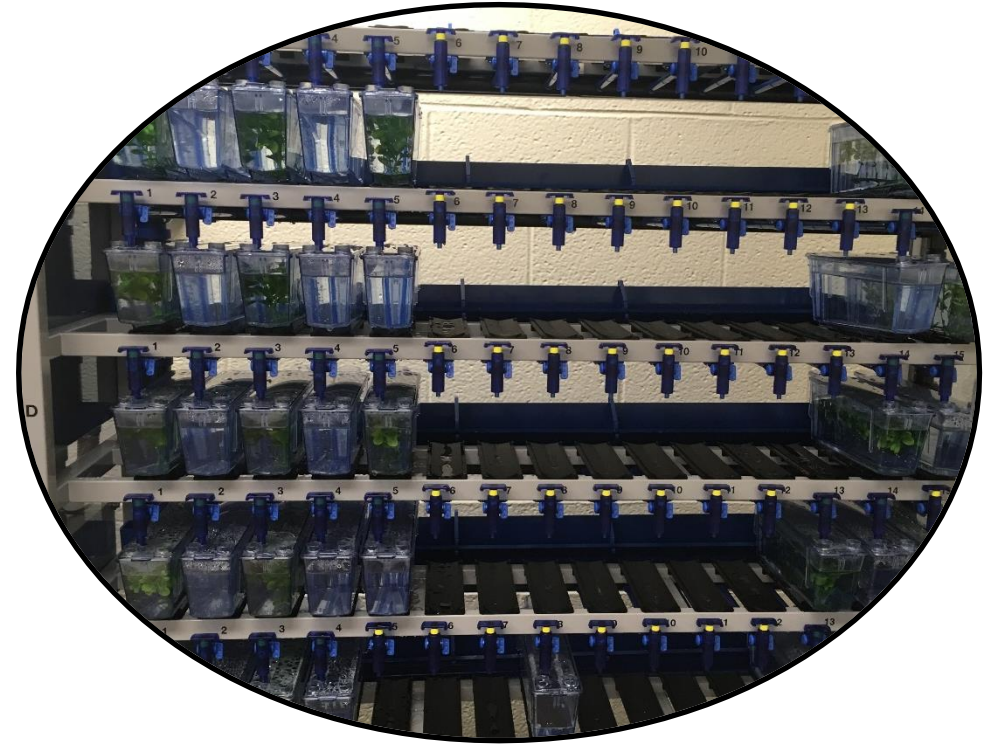


2018 Environmental Summary: Berryhill

ROOM CHARACTERISTICS:

- **System manufacturer:** Tecniplast
- **Start date:** January 2018
- **Aquaculture type:** Flow-through
- **# Systems:** 1
- **System size (# racks) :** 1
- **System volume (total liters) :** ~300
- **Standard tank size:** 1.1 liter
- **Tank color:** Clear, blue tint
- **Tank lid color:** Clear, blue tint
- **Component materials:** PVC, polycarbonate
- **Filtration type (solids):** Rotating drum (26 μ M)
- **Filtration type (biological):** Moving bed, Kaldnes biochips
- **Filtration type (chemical):** N/A
- **Water exchange rate (system volume) :** N/A
- **Water exchange rate (tank):** 2-4 changes/hour
- **Photoperiod:** 14H Light: 10H Dark
- **Mean light intensity (lux), front of tank:** NR
- **Mean light intensity (lux), rear of tank:** NR
- **Mean % difference in light intensity (lux), front to back of tank:** NR



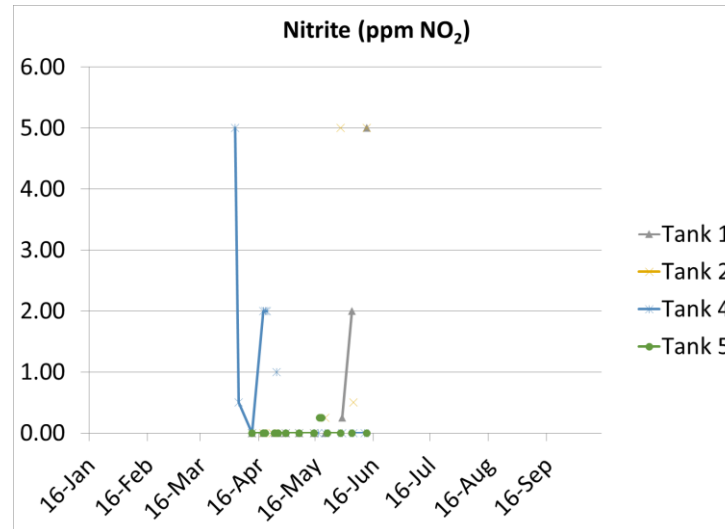
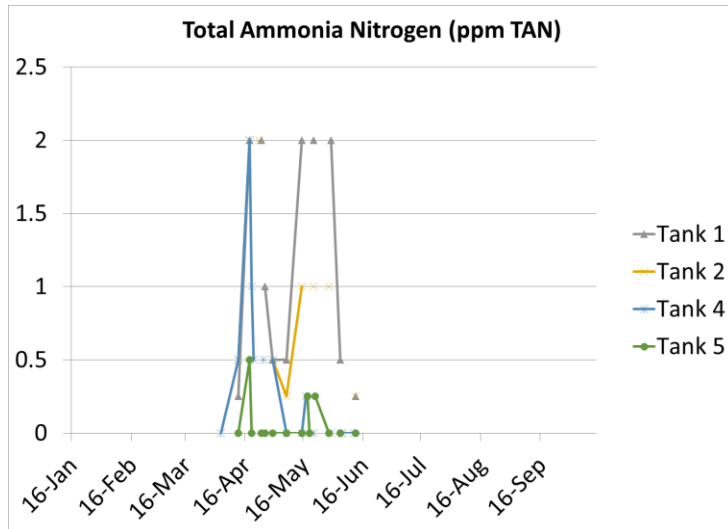
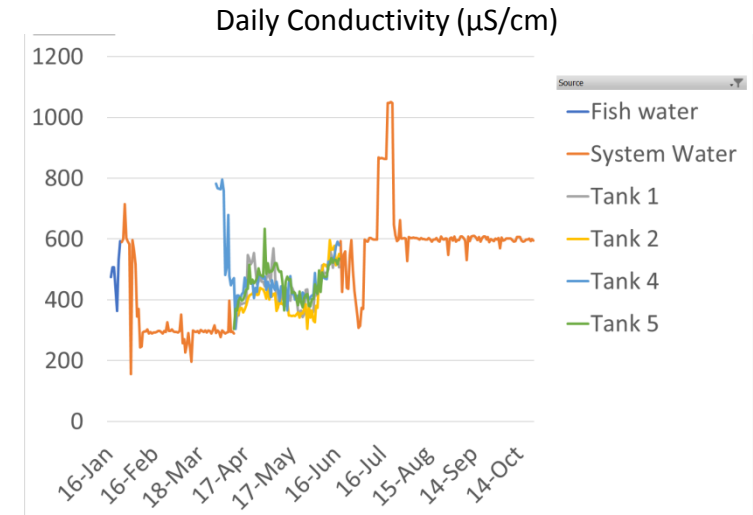
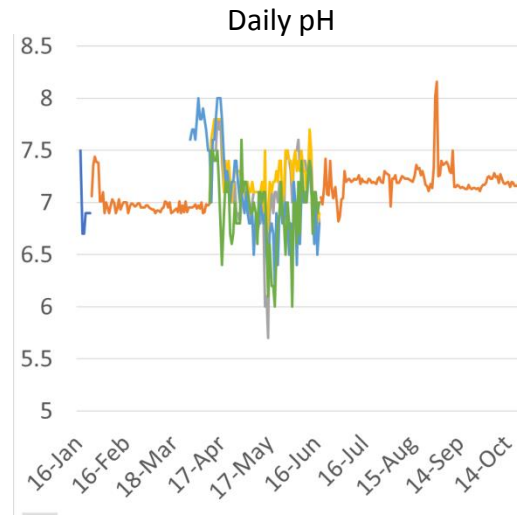
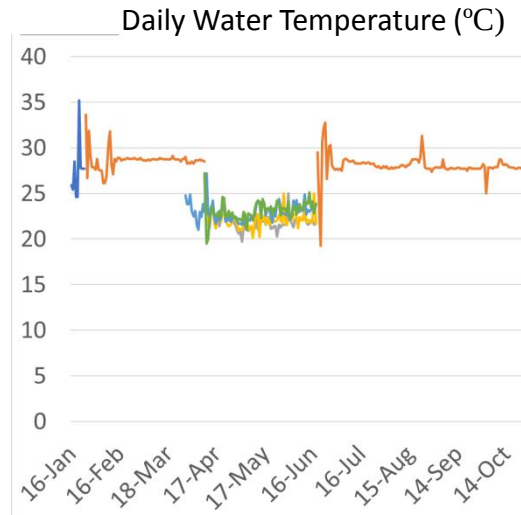
2018 Environmental Summary: Berryhill (Infection Study)

ROOM CHARACTERISTICS:

- **System manufacturer:** Aqua Culture, Ontario, Canada
- **Start date:** January 2018
- **Aquaculture type:** Static
- **# Systems:** 4
- **System size (# racks) :** N/A
- **System volume (total liters) :** 80L
- **Standard tank size:** 10 gallons
- **Tank color:** Clear
- **Tank lid color:** Black
- **Component materials:** Glass divided with nylon mesh
- **Filtration type (solids):** Sponge filters (Aquarium Technology, Inc. Decatur, GA)
- **Filtration type (biological):** N/A
- **Filtration type (chemical):** N/A
- **Water exchange rate (system volume) :** N/A
- **Water exchange rate (tank):** 90% once per week
- **Photoperiod:** 14H Light: 10H Dark
- **Mean light intensity (lux), front of tank:** NR
- **Mean light intensity (lux), rear of tank:** NR
- **Mean % difference in light intensity (lux), front to back of tank:** NR



2018 Environmental Summary: Berryhill



2018 ENVIRONMENTAL SUMMARY: ALL

NUTRITION AND FEEDING

- **Feed:** RGComplete
- **Size:** 1.5-12µm
- **Age of Application:** N/A
- **Feed Frequency:** fed to rotifers hourly
- **Amount/Feeding:** 11ml/day per 1 million rotifers collected
- **Ingredients:**
 - Seawater
 - Sodium Bicarbonate
 - Sodium Hydroxymethanesulfonate
 - Marine Microalgae
 - Citric acid
 - Ascorbic Acid
 - Larixinic Acid
 - Sodium Alginate

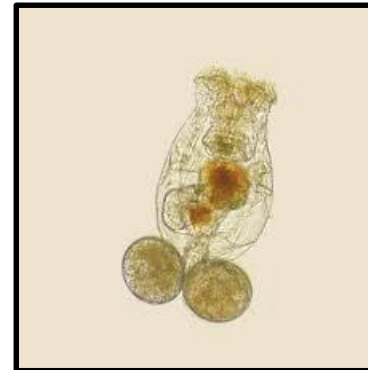
Guaranteed analysis:

- Crude Protein (min) 2.7%
- Crude Fat (min) 0.9%
- Crude Fiber (max) 0.2%
- Moisture (max) 89.0%
- Microalgal biomass (min) 4.4%
- Total Hufas (min) 64 mg/g d.w. (biomass)
- ARA (min) 2.1 mg/g d.w. (biomass)
- EPA (min) 18.1 mg/g d.w. (biomass)
- DHA (min) 41.9 mg/g d.w. (biomass)



2018 ENVIRONMENTAL SUMMARY: ALL NUTRITION AND FEEDING

- **Feed:** Rotifers, Type L (*Brachionus plicatilis*)
- **Size:** ~160 μm
- **Age of Application:** < 21 dpf
- **Feed Frequency:** constant availability 5-8 dpf; fed once per day 8-21 dpf
- **Amount/Feeding:** polyculture of ~250 rotifers per ml from 5-8 dpf; ~1,000 rotifers per fish 8-21 dpf



2018 ENVIRONMENTAL SUMMARY: ALL NUTRITION AND FEEDING

- **Feed:** Gemma Micro
- **Size:** 150 µm
- **Age of Application:** 21 - 30 dpf
- **Feed Frequency:** 1x/day
- **Amount/Feeding:** ~3-5% of body weight

GEMMA Micro zebrafish Detailed composition				
INGREDIENTS	Minerals	Vitamins	Fatty Acids	Amino Acids % of RMs
Fish meal	Calcium < 15,000 ppm	Vitamin C 1,000 ppm	N-3 40 mg/g	Arg 3.50 %
Lecithin	Phosphorus < 20,000 ppm	Vit. A 23,000 IJ/kg	N-3 % 19 % of fat	His 1.25 %
Wheat gluten	Sodium > 5,000 ppm	Vit. D 2,800 IJ/kg	ARA 1 mg/g	Lis 3.00 %
Dried Seaweed	Manganese 40 ppm	Vit. E (total) 750 ppm	DHA 23 mg/g	Leu 4.00 %
Fish oil	Potassium 3,321 ppm	Vit. B1 80 ppm	EPA 10 mg/g	Lys 3.90 %
Maize starch	Copper 10 ppm	Vit. B2 45 ppm	DHA:EPA ratio >2	Thr 1.50 %
Vitamins & Minerals	Iron 220 ppm	Vit. B3 110 ppm	N-6 6,6 mg/g	Trp 0.70 %
	Zinc 170 ppm	Vit. B5 160 ppm	N-6 % 5,3 % of fat	Val 2.20 %
	Selenium 1,7 ppm	Vit. B6 110 ppm		Met+Cys 2.05 %
	Iodine 5 ppm	Vit. B7 1,2 ppm		Phe+Tyr 3.50 %
		Vit. B9 9 ppm		Tau 3.00 %
		Vit. B12 770 µg/kg		
		Choline 3,000 ppm		
		Aetha 38 ppm		
COMPOSITION				
Protein 59 %				
Fat 14 %				
Ash 14 %				
Moisture 8 %				
Crude fiber 0.2 %				
Starch 1.9 %				
The values are indicative and may be subject to change				
version 1.0 (June 2017)				



2018 ENVIRONMENTAL SUMMARY: ALL NUTRITION AND FEEDING

- **Feed:** Gemma Micro
- **Size:** 300 µm
- **Age of Application:** >30 dpf
- **Feed Frequency:** 1x/day (2x/day >130 dpf)
- **Amount/Feeding:** ~3-5% of body weight

GEMMA Micro zebrafish Detailed composition				
INGREDIENTS	Minerals	Vitamins	Fatty Acids	Amino Acids % of RMs
Fish meal	Calcium < 15,000 ppm	Vitamin C 1,000 ppm	N-3 40 mg/g	Arg 3.50 %
Lecithin	Phosphorus < 20,000 ppm	Vit. A 23,000 IJ/kg	N-3 % 19 % of fat	His 1.25 %
Wheat gluten	Sodium > 5,000 ppm	Vit. D 2,800 IJ/kg	ARA 1 mg/g	Lis 3.00 %
Dried Seaweed	Manganese 40 ppm	Vit. E (total) 750 ppm	DHA 23 mg/g	Leu 4.00 %
Fish oil	Potassium 3,321 ppm	Vit. B1 80 ppm	EPA 10 mg/g	Lys 3.90 %
Maize starch	Copper 10 ppm	Vit. B2 45 ppm	DHA:EPA ratio >2	Thr 1.50 %
Vitamins & Minerals	Iron 220 ppm	Vit. B3 110 ppm	N-6 6,6 mg/g	Trp 0.70 %
	Zinc 170 ppm	Vit. B5 160 ppm	N-6 % 5,3 % of fat	Val 2.20 %
	Selenium 1,7 ppm	Vit. B6 110 ppm		Met+Cys 2.05 %
	Iodine 5 ppm	Vit. B7 1,2 ppm		Phe+Tyr 3.50 %
		Vit. B9 9 ppm		Tau 3.00 %
		Vit. B12 770 µg/kg		
		Choline 3,000 ppm		
		Aetha 38 ppm		
COMPOSITION				
Protein 59 %				
Fat 14 %				
Ash 14 %				
Moisture 8 %				
Crude fiber 0.2 %				
Starch 1.9 %				
The values are indicative and may be subject to change				
version 1.0 (June 2017)				

2018 ENVIRONMENTAL SUMMARY: ALL NUTRITION AND FEEDING

- **Feed:** Zeigler Zebrafish Diet
- **Size:** 1000 μm
- **Age of Application:** >110 dpf
- **Feed Frequency:** 1x/day (2x/day >130 dpf)
- **Amount/Feeding:** ~3-5% of body weight

INGREDIENTS

Fish Meal, Poultry By-Product Meal, Wheat Flour, Hydrolyzed Poultry Feather Meal, Soy Protein Concentrate, Fish Oil, Fish Protein Concentrate, Soy Protein Isolate, Grain Distillers Dried Yeast, Soy Lecithin, Monocalcium Phosphate, Yeast Culture, Vitamin A Acetate, Vitamin D3 Supplement, dl-Alpha Tocopheryl Acetate (Vitamin E Supplement), Vitamin B12 Supplement, Riboflavin-5-Phosphate, Niacin, d-Calcium Pantothenate, Menadione Sodium Bisulfite Complex (source of Vitamin K Activity), Folic Acid, Thiamine Mononitrate, Pyridoxine Hydrochloride, Biotin, Inositol, L-Ascorbyl-2-Polyphosphate (source of Vitamin C), Choline Chloride, Yeast Extract, Manganese Proteinate, Zinc Proteinate, Copper Proteinate, Calcium Iodate, Iron Proteinate, Cobalt Proteinate, Calcium Carbonate, Sodium Selenite, Calcium Propionate (Preservative), Astaxanthin, Zinc Oxide.



GUARANTEED ANALYSIS

Crude Protein Minimum 55.0%
Crude Fat Minimum 15.0%
Crude Fiber Maximum 1.5%
Moisture Maximum 12.0%
Ash Maximum 12.0%
Phosphorus Minimum 1.5%

CALCULATED NUTRIENT COMPOSITION

MINERAL CONCENTRATION

Calcium % 2.78
Phosphorous % 1.78
Potassium % 0.65
Sodium % 0.37
Magnesium % 0.13
Iron ppm 309.00
Zinc ppm 208.00
Manganese ppm 57.00
Copper ppm 40.00
Cobalt ppm 0.41
Iodine ppm 1.19

AMINO ACID CONCENTRATION (% OF TOTAL DIET)

Arginine 3.52
Lysine 3.53
Methionine 1.18
Cystine 0.75
Tryptophan 0.54
Histidine 1.22
Leucine 3.91
Isoleucine 2.39
Phenylalanine 2.19
Tyrosine 1.67
Threonine 2.16
Valine 2.77

VITAMIN CONCENTRATION

Vitamin A IU/g 30.24
Vitamin D3 IU/g 8.97
Alpha-Tocopherol IU/kg 434.50
Thiamine ppm 61.36
Riboflavin ppm 85.76
Niacin ppm 433.23
Pantothenic Acid ppm 184.81
Choline ppm 363.40
Pyridoxine ppm 58.66
Folic Acid ppm 18.21
Biotin ppm 1.12
Vitamin B12 Mcg/kg 146.98
Vitamin K ppm 17.86
Gross Energy Kcal/gm 3.89

2018 ENVIRONMENTAL SUMMARY: ALL

LEGEND

Physical and Chemical Parameters: (Testing modality, frequency of sampling) **Temperature** (Inline probes, continuous, value recorded daily). **pH** (inline probes, continuous, value recorded daily).

Conductivity (inline probes, continuous, value recorded daily). **Total Ammonia Nitrogen** (TAN) (API freshwater test kit, once weekly, stationary tanks only). **Nitrite** (API freshwater test kit, once weekly, stationary tanks only). **Nitrate** (NR, Flow-through system). **Hardness** (NR, Flow-through system), **Alkalinity** (NR, flow-through system). **UV Dose** (NR). **Dissolved Oxygen** (NR). **Carbon Dioxide** (NR). **Phosphate** (NR). **Light Intensity** (NR).